

WinContextDeploy

Post-image configuration and quick device diagnostics

Technician's manual

What the tool does, what it leaves to you,
and how to extend it

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Introduction

1.1 Why this tool exists

A freshly imaged Windows machine is not a finished machine. The image is the same for everyone, but the machine in front of you is not: it has a battery or it does not, it runs its applications locally or in a remote session, it belongs to a technician who needs a French interface or an English one.

Closing that gap by hand takes twenty minutes of clicking through the same settings dialogs, in the same order, and it takes them again on the next machine. Twenty minutes is not the problem. The problem is that the twentieth machine of the week gets a slightly different treatment from the first, and nobody can say afterwards which settings were actually applied.

WinContextDeploy is the answer to that: a technician runs it once, answers five or six questions, and it applies the settings that depend on the machine's context, opens the applications that need to be checked by eye, and prints a checklist of what was done, what was skipped, and what still has to be done by hand.

CAUTION

It does **not** install software. Applications are expected to arrive through your imaging pipeline. This tool confirms they are there and puts the ones a technician must eyeball in front of them.

1.2 The three ways to launch it

There is one launcher, `WinContextDeploy.cmd`, and it covers three situations.

`WinContextDeploy.cmd`

Run **in place** — correct for a `git clone`

`WinContextDeploy.cmd -Usb`

Copy to `%TEMP%`, run there, write the log back

`powershell -File src\Invoke-WcdConfiguration.ps1`

Direct, scriptable

Run in place is the default and the one to use after a `git clone`. Nothing is copied and nothing is deleted.

-Usb is for a USB key. PowerShell off removable media is slow and the key can be pulled mid-run, so the project is copied to a uniquely named folder under `%TEMP%`, run from there, and the run's history block is appended back to `log.txt` next to the launcher before the copy is removed.

Direct is how you drive it from another script: every prompt has a matching parameter, so `-NonInteractive` runs the whole thing without asking anything.

Add `-FR` or `-EN` to either `.cmd` form to force the interface language. Without one, it follows the system locale.

1.3 What this manual covers

- **Chapters 2 and 3** — how to run the tool, and what it does on its own.
- **Chapter 4** — what it deliberately leaves to you.
- **Chapter 5** — how the project is put together, in diagrams.
- **Chapter 6** — PowerShell, explained from nothing.
- **Chapter 7** — adding your own module.

- **Chapter 8** — reading the diagnostic and the logs when something goes wrong.

Quick start

2.1 Before you run it

Windows	Windows 10 or 11.
PowerShell	5.1 or later. Windows ships with it; nothing to install.
Administrator	Only the power settings, the BitLocker / TPM check and a domain join need it. The tool offers the UAC prompt itself and carries on perfectly well if you decline.
The manifest	Open <code>WinContextDeploy.psdl</code> and point it at the applications, printers and URLs of your environment. The shipped entries are generic examples.

CAUTION

Read the manifest before the first run on a real machine. The tool changes system settings: the taskbar, the regional separators, the power plan and the keyboard layout.

2.2 Running it

Double-click `WinContextDeploy.cmd`. A console window opens, the banner is drawn, and the questions start.



Every question works the same way: **Left** and **Right** move between the choices, **Enter** confirms. The highlighted option is the default, so holding Enter through all of them is a valid answer.

2.3 The menu questions

2.3.1 Windows system language

```
Windows system language
fr-CA = French Canadian Windows interface | en-US = American English
Use Left/Right to change, then Enter to confirm.
FR: fr-CA  EN: en-US
```

Sets the display language, the keyboard layout, the user culture and the home region. This is the language **Windows** will speak, and it is independent of the language the tool itself uses for its prompts.

2.3.2 Machine form factor

```
Machine form factor
Laptop = has a battery and a lid | Desktop = neither
```

```
Selects the power profile: battery and lid-close settings.
```

```
L: Laptop    D: Desktop
```

On a **Laptop**, the tool sets the screen timeout on battery, the screen timeout on AC, and makes closing the lid do nothing. On a **Desktop** the battery and lid steps do not apply, and the checklist says so rather than leaving them out.

2.3.3 Machine environment

```
Machine environment
```

```
Workstation = full local machine -> checks the locally installed applications
```

```
Citrix / VDI = thin endpoint -> its applications live in the remote session
```

```
W: Workstation    V: Citrix / VDI
```

This is what decides which applications are relevant. On a VDI endpoint the line-of-business applications live in the remote session, so checking for them locally would report a problem that is not one — the checklist marks them **Not Applicable** instead.

2.3.4 Open the configuration applications

```
Open Workstation configuration applications?
```

```
Opens the Application Targets declared in WinContextDeploy.psd1.
```

```
Answer Yes unless applications were already opened manually.
```

```
Y: Yes    N: No
```

Answer **No** if you have already opened everything by hand. The applications then appear on the checklist as manual steps rather than silently disappearing from it.

2.3.5 Machine identity

```
Machine identity
```

```
Serial: 5CG2141ABC (read-only, from firmware)
```

```
Change the computer name, or join the domain?
```

```
Neither is applied until the machine restarts.
```

```
Domain: corp.example.com
```

```
N: Neither    C: Computer name    D: Domain    B: Both
```

The serial is **displayed, never changed** — it is stamped into SMBIOS at the factory and no supported Windows API writes it. It is shown because it is what you read off the chassis label when deciding the name.

Answering **C** or **B** then asks for the name as free text. It is checked as you type it — 1 to 15 characters, no spaces or punctuation, not all digits, and not the name the machine already has — and a rejected name is re-asked immediately rather than failing at the end of the run.

NOTE

The domain options only appear when the manifest declares a `Domain.Name`. With none, the question is just the computer name.

CAUTION

The tool never restarts the machine. A reboot mid-run would destroy the checklist, the history log and the JSON report — which are the entire point of it. Neither the new name nor the domain membership takes effect until you restart, and the checklist carries a `Restart required` row to say so.

2.3.6 Engineering workstation

```
Engineering workstation? (offers the optional tools)
  Offers the extras declared Prompt in WinContextDeploy.psd1.
  Answer No for a standard machine.
Y: Yes   N: No
```

Answering **Yes** opens a numbered menu of the optional tools, where several can be picked at once by typing their numbers together — **13** selects the first and the third.

NOTE

This question only appears when the manifest declares at least one entry with `Prompt = $true`. With none, the tool skips it entirely.

2.4 The result: the final diagnostic

When every module has run, the tool prints its report twice. First grouped by module, which is the view that tells you whether the run itself went well:

```
=====
FINAL DIAGNOSTIC - BY MODULE
=====
[x] Config-Identity      OK      1 step(s)
[x] Config-Power        OK      5 step(s)
[x] Config-Decimal      OK      1 step(s)
[x] Config-TaskbarLeft  OK      2 step(s)
[x] Config-Language     OK      2 step(s)
[!] Config-Applications WARNING 9 step(s) Warnings: AppVpn
[x] Config-DeviceManager OK      1 step(s)
[x] Config-Disk         OK      2 step(s)
[x] Config-BitLocker    OK      2 step(s)
[x] Config-WindowsUpdate OK      2 step(s)
[x] Config-Network      OK      3 step(s)
```

Then as a flat checklist, which is the view you actually work from — one line per thing that had to happen on this machine:

```
=====
FINAL DIAGNOSTIC - BY STEP
=====
[x] Computer name      OK      Renamed to POSTE-01. Takes effect after
a restart.
[-] Domain membership  MANUAL  Not requested this run.
[-] Restart required   MANUAL  New computer name / domain membership
takes effect after a restart.

[x] Taskbar aligned left OK
[x] Display language   OK
[x] Keyboard layout    OK
[x] Decimal separator  OK
[x] Power options      OK
[x] Device Manager     OK
[x] Disk health        OK
[x] Free space         OK      412 GB free of 476 GB
[x] TPM readiness      OK      TPM present and ready.
[x] Drive encryption   OK      Protected: C: FullyEncrypted.
[x] Windows Update     OK      No failed update in the last 50
entries (30 days).

[x] Network adapters   OK
```

```

[x] Software Center      OK
[!] VPN client          WARNING    No matching process running (PanGPA, PanGPS).
                                -> Confirm VPN client is installed and started,
                                or mark the entry Optional.

[!] ERP client          ERROR     ERP client not found at C:\ProgramData\...
                                -> Update Applications['ERP client'].Target in
                                WinContextDeploy.psdl, or remove the entry.

[-] Citrix Workspace    N/A      Not applicable to the chosen Environment.
[-] Mail signature      MANUAL    Must be done manually.
[-] Wi-Fi               MANUAL    Must be done manually.
[-] Network drives      MANUAL    Must be done manually.

```

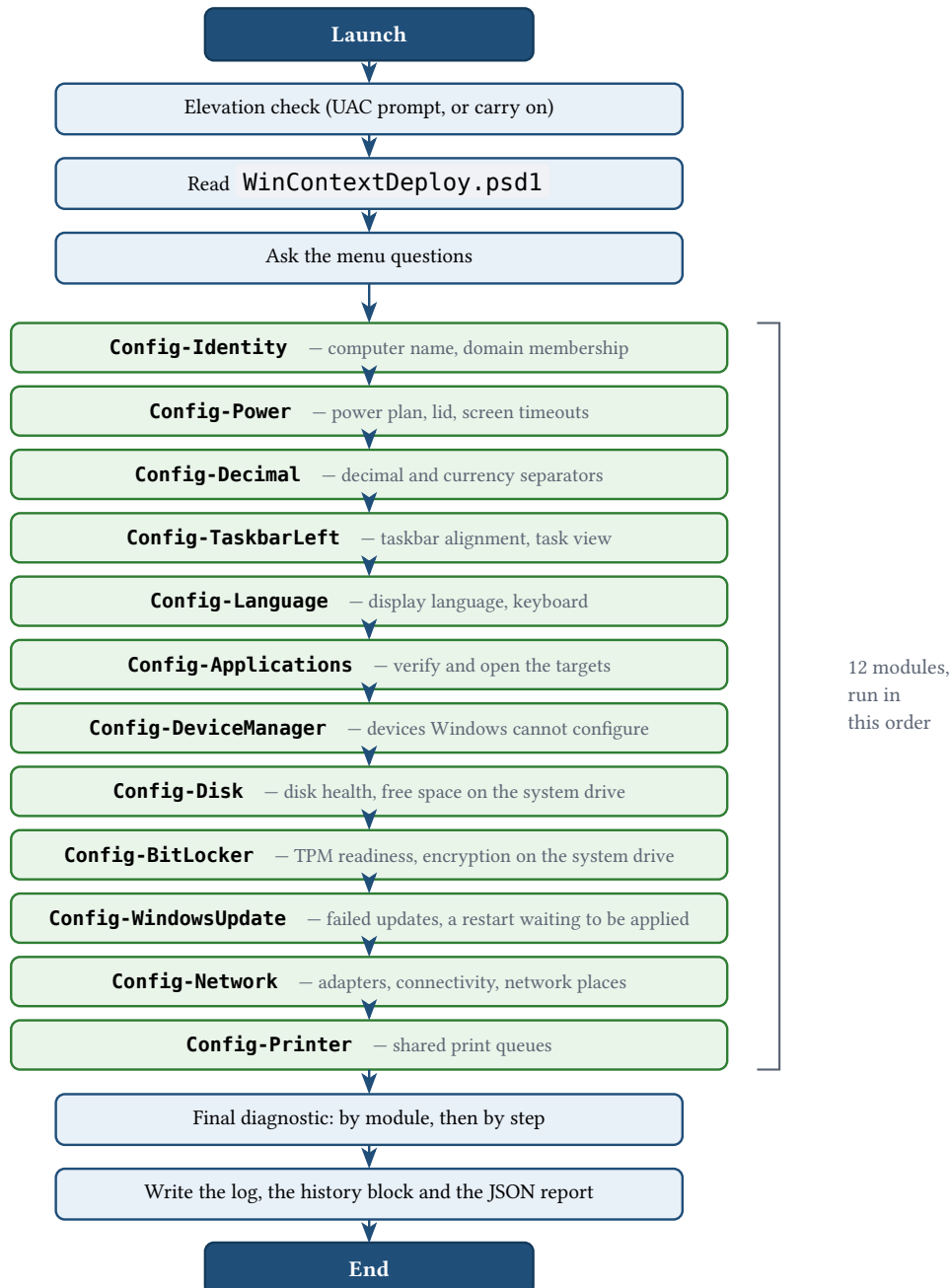
Summary: 11 OK, 1 warning(s), 1 error(s), 9 manual, 1 N/A.

The arrow lines are the point of the whole report. A failure that only says “*the specified file was not found*” tells a technician nothing; a failure that names the manifest key to edit tells a technician exactly what to do next.

What the tool does automatically

3.1 The full execution flow

From the double-click to the final report, the run always follows the same path. Your answers change what each module **does**, never the order they run in.



NOTE

A module that crashes does not stop the run. Its failure is recorded, the next module starts, and you still get a complete report at the end. The only thing a technician cannot use is a report that never appears.

3.2 What each module does

3.2.1 Config-Identity — computer name and domain membership

Naming the machine and joining it to the domain are the two steps whose omission is most obvious to the user and least obvious to the technician who has moved on. Both are driven from the machine-identity prompt, and both are skipped entirely when it is declined.

Choosing **Both** is a single `Add-Computer -NewName` call rather than a rename followed by a join — one operation, one restart. The two Steps then report the outcome of that one call.

The typed name is checked before any call is made, so a bad name is re-asked at the prompt instead of failing at the end of the run.

CAUTION

No credential ever goes in the manifest, the log, the history log or the JSON report. The manifest holds only the domain name and OU path; the technician types their own domain account into the standard Windows credential dialog at the moment of joining, and it exists for the length of one call.

The manifest is committed, shared, and frequently lives on a USB key. A plaintext join account in it is a domain-wide problem, not a local one.

TIP

The domain lives in the manifest:

```
Domain = @{
    Name   = 'corp.example.com'
    OUPath = 'OU=Workstations,DC=corp,DC=example,DC=com'
}
```

Leave `Name` empty and the domain option disappears from the prompt entirely, the same way an empty `Printers` array skips that module. `OUPath` is optional; omit it and the domain's own default container is used.

NOTE

Both cmdlets need Administrator, so an unelevated run attempts neither and opens no credential dialog: both Steps report as needing elevation, the same way the power Steps do.

`Get-Credential` cannot run unattended, so `-NonInteractive` can never join a domain. That is a documented limitation, not a bug. `-NewComputerName` and `-JoinDomain` skip the question but not the credential dialog.

Declining, or running `-NonInteractive`, leaves both rows on the checklist as Manual Steps rather than Not Applicable — a choice is not a filter, and a fleet-wide JSON report can then show which machines are still unnamed.

3.2.2 Config-Power — power plan, lid and screen timeouts

Sets the screen timeout to 10 minutes on battery and 15 on AC, makes closing the lid do nothing, and applies the active power scheme. On a `Desktop` the battery and lid steps do not apply.

One of the modules that need Administrator — `Config-BitLocker` and a domain join in `Config-Identity` are the others. Without it, the module attempts nothing and every step reports:

```
[!] Power options          WARNING    Screen timeout on AC: powercfg requires
                                Administrator.
                                -> Requires Administrator. Relaunch elevated
                                    to apply.
```

TIP

The sleep steps (`standby-timeout-*`) are deliberately switched off: they are blocked by Group Policy in most managed environments, so attempting them only produces noise. The commented rows in the module's step table are what to restore if that is not true for you.

3.2.3 Config-Decimal — decimal and currency separators

Forces the period as the decimal separator and the comma as the thousands separator, for both numbers and money. A French-Canadian Windows uses a comma for decimals by default, which quietly corrupts anything pasted into a spreadsheet built around the other convention.

3.2.4 Config-TaskbarLeft — taskbar alignment and task view

Aligns the taskbar to the left and hides the Task View button — Windows 11 defaults that most users on a fleet want changed once, not every time.

3.2.5 Config-Language — display language and keyboard

Applies the display language, the keyboard layout, the user culture, the system locale and the home region for the culture you chose.

NOTE

The system locale and the home region need Administrator. Without it, the module applies everything else and notes the two it could not do. Some Windows Store applications also only pick up the new language after a sign-out.

3.2.6 Config-Applications — verify and open the targets

The manifest-driven core of the tool. It walks the `Applications` list in `WinContextDeploy.psd1` in order and runs each entry according to its `Action` :

Launch	Starts an application: a <code>.lnk</code> , an <code>.exe</code> , or a command on <code>PATH</code> .
OpenFolder	Opens a folder in Explorer.
OpenUrl	Opens a URL in the default browser.
CheckProcess	Verifies a process is running. Launches nothing. <code>Target</code> is a list of process names, and any one of them counts.
CheckPath	Verifies a path exists. Launches nothing.
CheckWinget	Verifies a winget package id is installed. Launches nothing, and installs nothing. <code>Target</code> is the exact package id.

NOTE

`CheckWinget` exists because an install path moves between versions and between machines — a per-user MSIX lands somewhere quite different from a per-machine MSI — while the package id does not. winget itself is missing from LTSC and stripped images: the run probes for it once, reports that as a single warning,

and marks every `CheckWinget` entry as a Manual Step to verify by hand. One honest cause, and rows a technician can act on.

Adding, removing or reordering an application is a manifest edit. No code changes, no new module.

3.2.7 Config-DeviceManager — devices Windows cannot configure

Reads every Plug-and-Play device and reports the ones with a problem code. A missing driver on a freshly imaged machine is exactly what this catches, before the user finds it a week later.

Codes that resolve themselves — a pending reboot, a device not currently connected — are warnings. A missing or broken driver is an error.

3.2.8 Config-Disk — disk health and free space

A freshly imaged machine can still be sitting on a dying drive, or on a partition far smaller than the image expects. Neither shows up anywhere else in the run, and the technician usually finds out weeks later, from the user.

Two steps. `Disk health` reads the health each disk reports: `Healthy` passes, `Warning` is a warning, `Unhealthy` is an error that names the drive and its serial number so the right one gets pulled. A status the drive does not report at all is a warning rather than a silent pass — an unreadable disk is not a healthy one. `Free space` reads the system drive and reports the figure either way, so an OK row still tells you how much room the machine has.

Removable disks are left out of the health check. A technician's USB key is not the machine's disk, and it must never block a handover. Free space is read on the system drive only, for the same reason.

TIP

How much free space is enough is a fleet decision, so it lives in the manifest:

```
Disk = @{
    MinFreeGB = 20
}
```

Omit the key and the threshold is 20 GB. Below it the checklist warns and names the figure and the threshold; the remediation points back at `Disk.MinFreeGB`, so a threshold that is wrong for your fleet is one edit away.

NOTE

Wear percentage and drive temperature — `Get-StorageReliabilityCounter` — are deliberately not read. They are genuinely useful on refurbished machines, but they need Administrator and return nothing on plenty of consumer SATA and NVMe drives, so the step would silently report nothing on a large slice of any fleet.

3.2.9 Config-BitLocker — TPM readiness and drive encryption

Handing over an unencrypted laptop is the kind of mistake that is invisible on the day and expensive later. This module reports whether the system drive is protected and whether the TPM is in a state that would let it be.

Two steps. `TPM readiness` reads `Get-Tpm`: present and ready passes, anything else is a warning pointing at the firmware setting. `Drive encryption` reads `Get-BitLockerVolume` on the system drive: fully encrypted and protected passes, anything else warns.

An unprotected system drive is **always** a warning, and there is no manifest knob to silence it. A fleet that applies encryption by policy after enrolment will see the warning on a fresh machine, and that is the correct thing for the checklist to say.

NOTE

A drive still encrypting warns rather than passes, and the row names the percentage. Windows flips `ProtectionStatus` to `On` as soon as the protectors exist — long before the drive is actually encrypted — so a step keyed on that alone would print a green `OK` for a machine sitting at 47%. A machine handed over half encrypted is not a protected machine.

CAUTION

This module reports and nothing else. It never enables BitLocker and never reads or stores a recovery key. Turning encryption on is a security decision with a real lockout risk behind it, not a verification, and it belongs to whoever owns the fleet policy — not to a post-image checklist.

NOTE

Both cmdlets need Administrator, so an unelevated run reports both steps as needing elevation, exactly the way the power steps do. That is why this is not part of `Config-Disk : disk health` needs no elevation and works on every edition, so it keeps reporting normally on a non-elevated run instead of being dragged into a half-elevated module.

Windows Home ships without the BitLocker module at all. The step probes for the cmdlet rather than parsing the edition string — a stripped Professional image behaves the same way and would fool the string — and reports one row saying the edition does not support BitLocker, rather than crashing.

3.2.10 Config-WindowsUpdate — failed updates and a pending restart

A machine that failed an update during imaging, or that is sitting on an unapplied reboot, looks completely fine at handover and is not. Neither shows up anywhere else in the run.

Two steps, both reading state that is already on the machine. `Windows Update history` reads the local update history through the Windows Update Agent and warns about any recent entry that did not succeed, naming the update — the KB number is part of the title Windows records — and its `HRESULT`. `Restart pending` reads the

`RebootRequired` key: absent passes, present warns.

Only recent history counts: the last 50 entries, and only the last 30 days. A machine re-imaged over an older install carries history that has nothing to do with this deployment, and warning about it would be noise a technician learns to ignore.

NOTE

A restart is asked for **once**. If the machine was also renamed or joined to a domain, the same restart covers both, and the `Restart required` row says so rather than reading as though two were needed.

CAUTION

“Are updates pending” is deliberately not checked. That search asks Microsoft over the network, takes anywhere from 30 seconds to several minutes in a tool whose whole value is being fast, and can hang. Worse, a freshly imaged machine **always** has updates pending, because the image is weeks old — the step would warn on 100% of runs, which is exactly the alarm technicians learn to scroll past. “An update tried and failed” is the actionable signal; “updates exist” is not.

NOTE

The Windows Update Agent is absent or disabled on some managed and stripped images. Creating the COM object is wrapped: that reports one note saying the history could not be read, rather than crashing the module. Neither step needs Administrator.

3.2.11 Config-Network — adapters, connectivity and network places

Inventories the active adapters (the virtual ones — Bluetooth, loopback, VPN, hypervisor — are filtered out), pings the configured target **from the Wi-Fi adapter**, and triggers the user's "Refresh My Network Places" shortcut.

TIP

The ping source address is forced to the Wi-Fi adapter on purpose. With a cable plugged in, an ordinary ping would be answered over Ethernet and pass while the Wi-Fi is quietly broken.

A blocked ping is a warning, never a hard failure — many corporate networks drop ICMP to the internet. Point `Network.PingTarget` at your gateway or an internal host if that is your case.

3.2.12 Config-Printer — shared print queues

Connects the shared print-server queues listed in the manifest's `Printers` array, using the built-in `Add-Printer` cmdlet. Already connected is a no-op, so running the tool twice is safe, and an unreachable print server is a warning rather than a crash.

Leave `Printers = @()` and the module is skipped entirely; printers then stay on the checklist as a manual step.

What stays manual

Some things are deliberately not automated. Automating them badly is worse than not automating them: a step that half-works on some machines is a step nobody can trust on any machine.

They appear on the checklist anyway, marked `MANUAL`, so a technician cannot finish a machine having forgotten one.

Mail signature	Depends on the person, not the machine: their name, title and telephone number. There is nothing about the machine that can supply them.
Wi-Fi	Joining a network usually needs a credential or a certificate the tool has no business handling.
Network drives	Which shares a user gets is a directory decision. <code>Config-Network</code> refreshes network places, but it will not invent mappings.
Account sync	Signing into the account is the user's action, at their first login.
Windows desktop	Icon layout and personal shortcuts — a preference, not a configuration.
Browser favourites	Imported from the user's profile or pushed by policy, not by a local script.
Printers	Manual only when the manifest declares none . Fill in the <code>Printers</code> array and this becomes automatic.

NOTE

A manual step is not a failure and not a skipped step. The tool distinguishes three things that look alike:

- **MANUAL** — deliberately yours to do.
- **N/A** — does not apply to this machine's form factor or environment, so its absence is correct.
- **ERROR** — should have worked, and did not.

How the project is put together

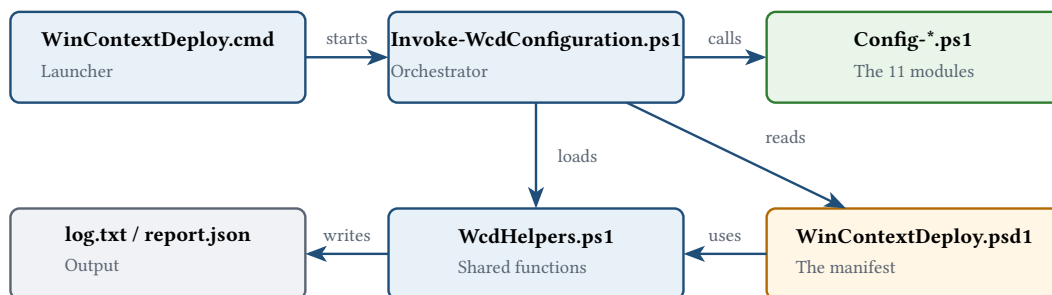
5.1 C4 diagram – the architecture

A C4 diagram describes a system at progressively closer zoom levels. Think of it as a map: level 1 is the satellite view of who uses what, level 2 shows the buildings inside.

5.1.1 Level 1 – context: who interacts with what



5.1.2 Level 2 – containers: the pieces inside



TIP

Notice which way the arrows point. The orchestrator reads the manifest and hands each module exactly what it needs; the modules return a Result and write to the log. Nothing reads back out of a module. That is what makes each one testable on its own — and why adding a module changes nothing else.

5.2 The folder tree

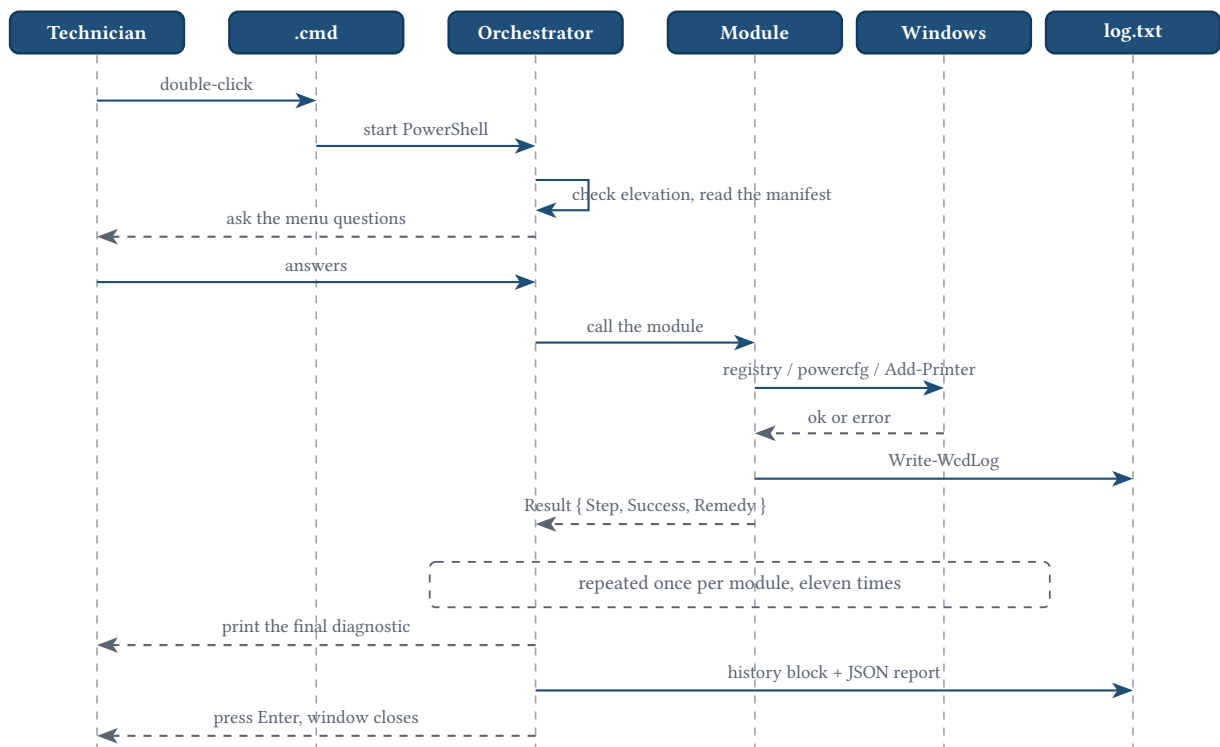
WinContextDeploy/

banner.txt	startup logo, editable
WinContextDeploy.cmd	the one launcher
WinContextDeploy.psd1	the manifest — edit this
src/	the code
Invoke-WcdConfiguration.ps1	orchestrator
WcdHelpers.ps1	shared functions
Config-*.ps1	one file per module
tests/	Pester tests, one file per module
docs/	this manual

<code>WinContextDeploy.psd1</code>	Every environment-specific value: application paths, printers, URLs, the ping target. This is the file you edit; the rest is the same everywhere.
<code>src/</code>	One file per module, plus the orchestrator and the shared helpers. Adding a module means adding a file here.
<code>tests/</code>	One Pester test file per module. A new module gets a new test file.
<code>docs/</code>	This manual, as Typst source and as a built PDF.

5.3 Sequence diagram — how the pieces talk

This is one run, in the order things actually happen.



NOTE

The module never touches the screen. It returns a `Result` and raises progress events; the orchestrator decides what that looks like. That is exactly why a module can be tested with no console attached at all.

PowerShell, explained simply

You do not need to be a developer to extend this tool. You need four ideas, and this chapter covers all four with examples taken from the code you already have.

6.1 What PowerShell is

Windows has a graphical interface and a text one. Clicking through the Settings app is fine for one machine. It is unbearable for twelve a week, and — the part that matters — it leaves no record of what was actually done.

PowerShell is the text interface. You write down the steps once, in a file, and Windows performs them the same way every time. That file **is** the record.

A PowerShell file has the extension `.ps1`. That is all a “script” is.

6.2 The four building blocks

6.2.1 Variables — remember a value

A variable is a labelled box. It starts with `$`, you put something in it, and you use the label afterwards instead of repeating the value.

```
# Store the registry key we are about to change
$intlPath = 'HKCU:\Control Panel\International'

# Use it twice without retyping it
Set-WcdRegistryValue -Path $intlPath -Name 'sDecimal' -Value '.'
Set-WcdRegistryValue -Path $intlPath -Name 'sThousandSep' -Value ','
```

If the key ever moves, you change one line, not two. On a longer script, not twenty.

6.2.2 Conditions — do something only sometimes

`if` runs a block only when something is true. This is how the tool adapts to the machine in front of it, and it is the whole idea behind the Form Factor question:

```
if ($FormFactor -eq 'Laptop') {
    # Only a laptop has a lid to close
    Invoke-WcdPowerCfg '/setacvalueindex' 'SCHEME_CURRENT' 'SUB_BUTTONS' 'LIDACTION' '0'
}
```

`-eq` means “equals”. The others read the same way: `-ne` not equal, `-gt` greater than, `-lt` less than, `-match` matches a pattern.

6.2.3 Functions — name a piece of work

A function is a named block you can call from anywhere. Every module in this project is a function.

```
function Set-WcdDecimalConfiguration {
    param(
        [string]$LogPath      # what the caller passes in
    )

    # ... the work ...
```

```

    return [pscustomobject]@{ Step = 'DecimalAndCurrency'; Success = $true }
}

# Calling it
$result = Set-WcdDecimalConfiguration -LogPath 'C:\temp\log.txt'

```

`param(...)` declares what the caller may pass. `return` hands something back.

6.2.4 try / catch — the safety net

This is the one that makes the whole tool usable. `try` attempts something; `catch` runs only if it failed, instead of the whole script stopping.

```

try {
    Set-WcdRegistryValue -Path $intlPath -Name 'sDecimal' -Value '.'
    $success = $true
} catch {
    # $_ is whatever went wrong
    $note = $_.Exception.Message
    $success = $false
}

```

TIP

This is why one locked registry key does not cost you the whole run. The module records the failure, returns, and the next module starts. A technician gets a full report of a partly successful run — which is far more useful than a script that stopped at step three with a red wall of text.

6.3 The anatomy of a module

`Config-Decimal` is the simplest module in the project: one step, no dependencies. Every other module has the same five zones — only zone 4 differs.

Zone 1 — Header	what the module does, its parameters, what it returns
Zone 2 — Function and parameters	<code>function Set-Wcd<X>Configuration { param(...) }</code>
Zone 3 — Initialisation	resolve the log path, name the module
Zone 4 — try / catch — the actual work	do it, raise progress events, log the outcome
Zone 5 — Return	<code>[pscustomobject]@{ Step; Success; Error; RemedyKey }</code>

Here is the real module, with the zones marked:

```

# ----- ZONE 1
# Config-Decimal.ps1 - forces the decimal and currency separators to a period.
# Entry point: Set-WcdDecimalConfiguration. Requires WcdHelpers.ps1.

function Set-WcdDecimalConfiguration {
    [CmdletBinding()]

```

```

param(
    [string]$LogPath,                # where to write the log
    [scriptblock]$ProgressCallback   # how to report progress
)

# ----- ZONE 3
$resolvedLogPath = Resolve-WcdLogPath -CandidatePath $LogPath
$intlPath = 'HKCU:\Control Panel\International'
$moduleName = 'Config-Decimal'

# ----- ZONE 4
try {
    # Tell the orchestrator this step is starting
    Invoke-WcdProgressCallback -ProgressCallback $ProgressCallback `
        -ModuleName $moduleName -StepKey 'DecimalAndCurrency' -Event 'Start'

    # The actual work
    Set-WcdRegistryValue -Path $intlPath -Name 'sDecimal' -Value '.'
    Set-WcdRegistryValue -Path $intlPath -Name 'sThousandSep' -Value ','
    Set-WcdRegistryValue -Path $intlPath -Name 'sMonDecimalSep' -Value '.'
    Set-WcdDecimalThreadCulture -DecimalSeparator '.' -ThousandsSeparator ','

    Write-WcdLog -Path $resolvedLogPath -Level 'INFO' `
        -Message 'Regional: decimal and currency separators forced to a period.'

    Invoke-WcdProgressCallback -ProgressCallback $ProgressCallback `
        -ModuleName $moduleName -StepKey 'DecimalAndCurrency' `
        -Event 'Finish' -Kind 'success'

    # ----- ZONE 5
    return [pscustomobject]@{
        Step = 'DecimalAndCurrency'
        Success = $true
        LogPath = $resolvedLogPath
        Error = ''
    }
} catch {
    # Something went wrong: note it, name the fix, and carry on
    $note = $_.Exception.Message
    $remedyKey = 'RegistryWriteFailed'
    if ($note -match 'access is denied|unauthorized') {
        $note = 'Registry key locked by GPO or access denied.'
        $remedyKey = 'RegistryGpo'
    }

    Write-WcdLog -Path $resolvedLogPath -Level 'ERROR' -Message "Regional: $note"
    Invoke-WcdProgressCallback -ProgressCallback $ProgressCallback `
        -ModuleName $moduleName -StepKey 'DecimalAndCurrency' `
        -Event 'Finish' -Kind 'error'

    return [pscustomobject]@{
        Step = 'DecimalAndCurrency'
        Success = $false
        LogPath = $resolvedLogPath
        Error = $note # the raw detail, for the log
        RemedyKey = $remedyKey # what the technician should do
    }
}

```

```
}  
  }  
}
```

TIP

Understand this module and you understand all eleven. `Config-Power` has six steps instead of one and `Config-Language` calls different cmdlets, but the shape never changes.

6.4 A word about `RemedyKey`

Notice what the `catch` block returns. It keeps the raw error **and** names a remedy — but it does not write the remediation sentence itself.

That is deliberate. The sentence is user-facing text, so it lives in the translation tables in `Invoke-WcdConfiguration.ps1`, in both languages, next to every other string a technician reads. The module only says **which** problem this is; the orchestrator decides how to say it, and in what language.

Adding a new module

7.1 First: does it need to be a module at all?

Most of what a technician wants to add is **opening or checking an application**, and that needs no code at all — it is one entry in `WinContextDeploy.ps1`:

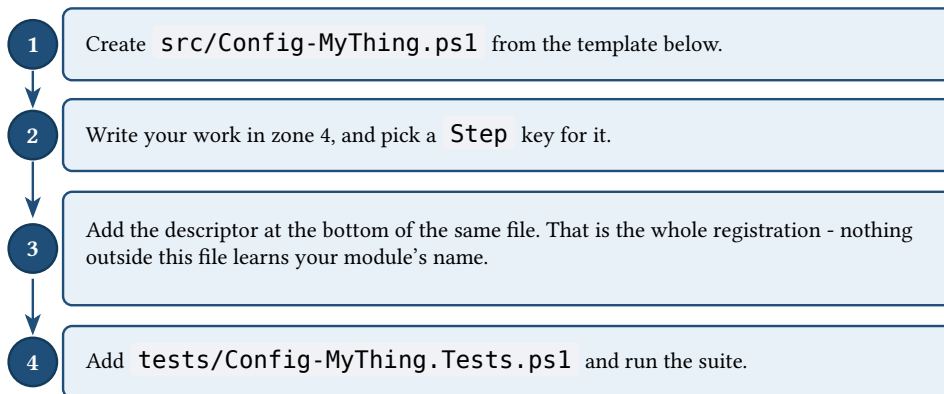
```
@{
    Step    = 'AppTimeTracking'
    Name    = 'Time tracking portal'
    Action  = 'OpenUrl'
    Target  = 'https://time.example.com/'
}
```

That entry gets a progress bar, a checklist row, a log line, an entry in the JSON report and a remediation sentence when it fails — for free, because `Config-Applications` already does all of that for every entry in the list.

CAUTION

Write a module only when the manifest genuinely cannot express what you need: a registry change, a Windows setting, an inventory check. If the answer is “open this” or “check this exists”, it is a manifest edit and you are done.

7.2 The four steps



7.3 The template

Copy this into `src/Config-MyThing.ps1` and replace `MyThing` and `MyStep` with your own names.

```
# Config-MyThing.ps1 - one line saying what this module is for.
# Entry point: Set-WcdMyThingConfiguration. Requires WcdHelpers.ps1.

function Set-WcdMyThingConfiguration {
    <#
    .SYNOPSIS
        One sentence: what this module does.

    .DESCRIPTION
        A paragraph: how it does it, and anything a reader would otherwise have
```

to work out from the code - what needs Administrator, what is deliberately left out, what is safe to run twice.

.PARAMETER LogPath

Full path to the log file. Resolved automatically when omitted.

.PARAMETER ProgressCallback

Scriptblock invoked at the start and end of each step.

.OUTPUTS

[pscustomobject[]] with Step, Success, Error and, on a failure, RemedyKey.

.EXAMPLE

Set-WcdMyThingConfiguration -LogPath 'C:\temp\log.txt'

```
#>
[CmdletBinding()]
param(
    [string]$LogPath,
    [scriptblock]$ProgressCallback
)

$resolvedLogPath = Resolve-WcdLogPath -CandidatePath $LogPath
$moduleName = 'Config-MyThing'
$results = @()

try {
    Invoke-WcdProgressCallback -ProgressCallback $ProgressCallback `
        -ModuleName $moduleName -StepKey 'MyStep' -Event 'Start'

    # -----
    # YOUR WORK GOES HERE
    # -----

    Write-WcdLog -Path $resolvedLogPath -Level 'INFO' -Message 'MyThing: done.'
    Invoke-WcdProgressCallback -ProgressCallback $ProgressCallback `
        -ModuleName $moduleName -StepKey 'MyStep' -Event 'Finish' -Kind 'success'
    $results += [pscustomobject]@{ Step = 'MyStep'; Success = $true; Error = '' }
} catch {
    Write-WcdLog -Path $resolvedLogPath -Level 'ERROR' `
        -Message ('MyThing: {0}' -f $_.Exception.Message)
    Invoke-WcdProgressCallback -ProgressCallback $ProgressCallback `
        -ModuleName $moduleName -StepKey 'MyStep' -Event 'Finish' -Kind 'error'
    $results += [pscustomobject]@{
        Step      = 'MyStep'
        Success    = $false
        Error      = $_.Exception.Message
        RemedyKey  = 'MyStepFailed' # add the sentence to both $T tables
    }
}

return $results
}
```

TIP

The template raises its own Finish event because it has one step. A module with several — Config-Disk, Config-BitLocker and Config-Network all have — should call

`Complete-WcdProgressStep` instead: it reads the Result your module just recorded and picks the right progress kind, so a dozen branches never repeat the mapping.

```
Complete-WcdProgressStep -ProgressCallback $ProgressCallback `
    -ModuleName $moduleName -StepKey 'MyStep' -Results $results
```

7.3.1 Invoke-WcdStep , and when not to use it

The template above writes twelve lines around the one line that does the work: two progress events, two log lines, a try/catch, and the Result literal twice. `Invoke-WcdStep` in `WcdHelpers.ps1` owns all of that, so the step becomes the work plus the sentences that describe it:

```
$results += Invoke-WcdStep -Module $moduleName -Key 'MyStep' `
    -LogPath $resolvedLogPath -ProgressCallback $ProgressCallback `
    -SuccessLog 'MyThing: done.' `
    -FailureLabel 'My thing' -FailureRemedy 'MyStepFailed' `
    -Action { <the work> }
```

The Action returns nothing when the step simply worked. When it has more to say, it returns a hashtable that is merged over the Result — that one mechanism covers every case a step has:

The Action returns

nothing

```
@{ Severity = 'WARNING'; Error = '...'; RemedyKey = '...' }
```

```
@{ Severity = 'ERROR'; Error = '...' }
```

```
@{ Applied = $true }
```

```
@{ RemedyKey = '...'; RemedyArgs = @('C:') }
```

```
@{ Log = '...' }
```

What it means

It worked. Anything the Action writes to the pipeline that is not a hashtable is ignored, so a step calling a chatty command still reads as a success.

The step did not run, and here is why. Reported, not failed — this is how `Config-Power` handles an unelevated run.

A severity read off the machine rather than earned by a throw. `Config-Disk` and `Config-BitLocker` decide this way.

An extra field the Diagnostic reads. `RebootPending` is the other one.

Format arguments for the remediation sentence.

Write this instead of `-SuccessLog`. Not part of the Result.

An Action that throws produces a failed Result carrying `-FailureRemedy`. When a step's failures are worth telling apart — a registry key locked by Group Policy is a different sentence to the technician than a write that

merely failed — pass `-OnFailure` a scriptblock that classifies the error, as `Config-TaskbarLeft` does for both of its steps at once.

CAUTION

Do not reach for it by reflex. `Config-Applications` and `Config-Printer` loop over manifest entries and decide mid-step which of several Results to record; `Config-Network` branches on what the adapters reported. Written out, those read fine. Squeezed through a runner, they need an escape hatch — and a runner that half its callers escape from is not a runner.

Ten of the twelve modules are deliberately left as they are. Convert one when it is genuinely the simple case, not to be consistent.

7.4 Declaring it

The orchestrator finds your module by globbing `src/Config-*.ps1`. It does not hold a list of modules, a dispatch table, a progress plan or a checklist layout, so there is nothing to add to any of them. Your module says what it is, once, at the bottom of its own file:

```
function Get-WcdMyThingDescriptor {
    <#
    .SYNOPSIS
        Declares what Config-MyThing contributes to the run.

    .PARAMETER ExecutionOptions
        Resolved run options: Language, FormFactor, Environment, OpenApps,
        OptionalTools, NewComputerName and JoinDomain.

    .PARAMETER Config
        The imported manifest.

    .PARAMETER Translations
        The active $T table, for the checklist row labels.

    .OUTPUTS
        [pscustomobject] with Name, Order, RowOrder, Steps, Rows and Invoke.

    .EXAMPLE
        Get-WcdMyThingDescriptor -ExecutionOptions $options -Config $config -
Translations $T
    #>
    [Diagnostics.CodeAnalysis.SuppressMessageAttribute('PSReviewUnusedParameter', '',
        Justification = 'Uniform descriptor signature; the orchestrator calls every
Module identically.')]
    [CmdletBinding()]
    param(
        [pscustomobject]$ExecutionOptions,

        [hashtable]$Config,

        [hashtable]$Translations
    )

    return [pscustomobject]@{
        Name      = 'Config-MyThing'
        Order     = 125             # run order
        RowOrder  = 125             # position of its rows in the checklist
    }
```

```

Steps    = @(
    @{ Key = 'MyStep'; Label = 'My thing' }
)
Rows     = @(
    @{ Label = $Translations.Checklist.MyThing; Steps = @('MyStep') }
)
Invoke   = {
    param($ctx)

    Set-WcdMyThingConfiguration -LogPath $ctx.LogPath -ProgressCallback
$ctx.ProgressCallback
    }
}

```

The function name is derived from the file name: `Config-MyThing.ps1` must export `Get-WcdMyThingDescriptor`. Get it wrong, or leave a field out, and the run stops at startup naming what is missing — `Test-WcdModuleDescriptor` in `WcdHelpers.ps1` holds the contract, and `tests/ModuleDescriptor.Tests.ps1` runs every module through it.

7.4.1 What each field is for

Field	What it does
Name	Must match the file name. The by-module diagnostic shows it.
Order	Run order. Pick a gap between two existing modules.
RowOrder	Where your rows land in the technician's checklist. Independent of <code>Order</code> : the checklist is grouped for reading, not for running.
Steps	Every Step key your module can report, with its label. A Step carrying <code>Planned = \$false</code> is declared but not counted — that is how a Step filtered out by Form Factor, or one you only report on failure, keeps its label without making the progress bar overshoot. An empty <code>Steps</code> array means “nothing to do this run”: the module is skipped, and its rows still appear.
Rows	Checklist rows. <code>@{ Label; Steps }</code> takes its status from the Results those Steps produced. <code>@{ Label; Kind; Detail }</code> states a fixed fact instead. Add <code>MissingKind</code> / <code>MissingDetail</code> to say what a row means when nothing reported, or <code>OmitWhenMissing = \$true</code> to drop the row entirely in that case.
Invoke	<code>param(\$ctx)</code> , returns your Results. <code>\$ctx</code> carries <code>ExecutionOptions</code> , <code>Config</code> , <code>Translations</code> , <code>DomainTarget</code> , <code>Elevated</code> , <code>LogPath</code> and <code>ProgressCallback</code> — everything a module is allowed to need, so no module reaches into the orchestrator's variables.

TIP

The one thing still outside your file is the row label. `$Translations.Checklist.MyThing` has to exist in **both** `$T` tables in `src/Invoke-WcdConfiguration.ps1`, or the row renders blank in one language. `tests/Translations.Tests.ps1` fails when you add it to one table and forget the other.

You can skip that and write the label as a literal string in the descriptor — but then it says the same thing to a French technician as to an English one.

7.5 Testing it

Every module has a test file, and yours should too. The pattern is always the same: dot-source the helpers and the module, mock whatever touches the machine, and assert on the Results.

```
Describe 'Config-MyThing' {
    BeforeAll {
        $srcDir = Join-Path (Split-Path $PSScriptRoot -Parent) 'src'
        . (Join-Path $srcDir 'WcdHelpers.ps1')
        . (Join-Path $srcDir 'Config-MyThing.ps1')
    }

    It 'succeeds when the work goes through' {
        Mock -CommandName 'Set-WcdRegistryValue' {}

        $results = @(Set-WcdMyThingConfiguration -LogPath (Join-Path $TestDrive
'log.txt'))

        $results[0].Step | Should -Be 'MyStep'
        $results[0].Success | Should -BeTrue
    }

    It 'reports a failure without throwing' {
        Mock -CommandName 'Set-WcdRegistryValue' { throw 'Access is denied' }

        $results = @(Set-WcdMyThingConfiguration -LogPath (Join-Path $TestDrive
'log.txt'))

        $results[0].Success | Should -BeFalse
        $results[0].Error | Should -Match 'denied'
    }
}
```

TIP

Mock anything that touches the real machine — the registry, powercfg, Add-Printer, Start-Process. A test that changes the machine it runs on is a test nobody will run twice.

Run the suite with `Invoke-Pester .\tests\*.Tests.ps1 -Output Detailed`, or by double-clicking `Lancer-Tests-Pester.cmd`.

Troubleshooting

8.1 Reading the diagnostic statuses

Every row of the final checklist carries one of five statuses. Telling them apart is most of what troubleshooting is.

Status	Colour	What it means
[x] OK	Green	The step did what it was supposed to. It may still carry a note — an Optional application that is simply not installed says so here.
[!] WARNING	Yellow	Something worth your attention that is not a failure: a VPN client not running, a print server out of reach, a power step that needed Administrator.
[!] ERROR	Red	The step should have worked and did not. Read the remediation on the same line; the raw detail is in the log.
[-] MANUAL	Yellow	Deliberately not automated. Yours to do — see chapter 4.
[-] N/A	Grey	Does not apply to this machine's form factor or environment. Its absence is correct, not a problem.

TIP

The line under a warning or an error is the one to act on. It names the fix, and where a manifest entry is at fault, the exact key to edit:

```
[!] ERP client          ERROR    ERP client not found at C:\ProgramData\...
-> Update Applications['ERP client'].Target in
    WinContextDeploy.psdl, or remove the entry.
```

8.2 Reading the logs

Three files can come out of a run, and they answer different questions.

The run log src/log.txt	Everything this run did, line by line, with the raw error text. Override the location with <code>-LogPath</code> . This is where you look when the remediation on screen is not enough.
The history log -HistoryLogPath	One block appended per machine — name, serial number, model, user, the whole run log and the final diagnostic. One file becomes the record of every machine you configured. The <code>-Usb</code> launcher points this at the key automatically.
The JSON report -ReportPath	The same result, machine-readable, for collecting across a fleet. Carries a <code>schemaVersion</code> so a collector can version against it.

Each log line is a timestamp, a level, and a message:

```
2026-08-29 09:14:23 [INFO] Power: screen timeout on AC set to 15 min.
2026-08-29 09:14:23 [INFO] Power: active scheme applied.
2026-08-29 09:14:24 [INFO] Regional: decimal and currency separators forced to a period.
2026-08-29 09:14:25 [ERROR] Taskbar align: Registry key locked by GPO or access denied.
2026-08-29 09:14:26 [INFO] Display language: fr-CA applied.
2026-08-29 09:14:27 [WARNING] Set-WinSystemLocale not applied (Access denied).
```

NOTE

An [ERROR] line does not mean the tool crashed. It never stops on a failed step: the error is recorded so you can deal with it afterwards, and the run continues to the end. A report you can read beats a run that stopped at step three.

8.3 Common problems

Symptom	Likely cause	What to do
The window closes immediately	PowerShell execution policy	Launch through <code>WinContextDeploy.cmd</code> , never the <code>.ps1</code> directly. The launcher passes <code>-ExecutionPolicy Bypass</code> for exactly this reason.
Every power step says it needs Administrator	You declined the UAC prompt, or the account has no rights to give	Nothing is broken — everything else applied. Relaunch elevated to finish the power settings and read the encryption status, or accept them as they are.
The domain join failed, or the credential dialog was cancelled	Domain unreachable, the account cannot join machines, or a bad <code>OUPath</code>	Nothing else in the run is affected. Fix the cause and re-run with <code>-JoinDomain</code> , or join the machine by hand — the checklist keeps the row.
This edition does not support BitLocker	Windows Home, or an image with the BitLocker module stripped out	Not a fault of the tool. Confirm the machine is meant to ship on that edition, or reimage with one that supports encryption.
Registry key locked by in GPO the log	Group Policy owns that setting	Expected on a managed machine. The change has to be made in Group Policy; the tool cannot and should not fight it.
An application says it was not found	The manifest points at a path that does not exist here	The remediation names the key: fix <code>Applications['<name>'].Target</code> , or remove the entry if that application is not part of this image.
Module not found in the diagnostic	A <code>Config-*.ps1</code> is missing or misnamed	Check that the file exists in <code>src/</code> and that its name matches the <code>File</code> value in <code>\$modules</code> .
Free space warns on a machine that is fine	The threshold does not suit this fleet	The remediation names the key: raise or lower <code>Disk.MinFreeGB</code> in the manifest. It defaults to 20 GB, which is generous on a 128 GB endpoint.
The connectivity test fails on a working network	The network drops ICMP to the internet	Point <code>Network.PingTarget</code> in the manifest at your gateway or an internal host.
The history export failed after elevating	<code>-HistoryLogPath</code> is on a network share	UAC opens a new logon session, which drops mapped drives. Use a local or

Symptom	Likely cause	What to do
		USB path, or run elevated from the start. The tool warns before elevating when it sees a UNC path.
A printer never connects	The print server is unreachable, or the queue name is wrong	Check <code>Printers[].ConnectionName</code> against what the print server publishes. <code>Name</code> must match the published queue name for the idempotency check to work.

8.4 Getting more detail

Every function in `src/` carries proper help. When you want to know what something does or what it returns, ask PowerShell rather than reading the source:

```
Get-Help Set-WcdPowerConfiguration -Full
Get-Help Get-WcdApplicationTarget -Examples
```

And when a change to the code is what you need, the test suite is the fastest way to find out whether it worked:

```
Invoke-Pester .\tests\*.Tests.ps1 -Output Detailed
```